

Week 3 Task: Supply Chain Analytics Challenge

Tools: PYTHON + Power BI	Duration: 7 Days
Dataset: Supply Chain Dataset.csv	

[Click to download](#)

Objective

This week you will work on a real-world supply chain dataset and learn how data analysts handle complex business data.

Focus of this week:

- How to clean real-world messy data using Python
- How to analyze delivery performance and cost efficiency
- How to create business dashboards in Power BI
- How to generate insights from logistics data

About Dataset

- Shipment details
- Country and vendor information
- Product and item details
- Delivery timelines
- Cost and freight information

You will use this dataset to analyze:

- Delivery performance
- Cost efficiency
- Shipment patterns
- Vendor performance

Your Tasks

PART 1: Data Analysis with Python

Task 1: Load Dataset

- Load the dataset using Pandas
- Display first 10 rows
- Check shape and column names

Task 2: Data Cleaning

Perform the following cleaning steps:

1. Convert all date columns into proper datetime format
2. Identify missing values
3. Handle missing values (fill or drop based on logic)
4. Convert numeric columns (if stored as text):
 - Freight Cost
 - Weight
 - Unit Price

Task 3: Exploratory Data Analysis (EDA)

Perform analysis on:

Shipment Analysis

- Most common shipment mode
- Total shipments per country

Delivery Performance

- Calculate delivery delay (Scheduled vs Delivered Date)
- Find average delivery time
- Identify delayed shipments

Cost Analysis

- Total Freight Cost
- Average Line Item Value
- Total Insurance Cost (if available)

Task 4: Group-Based Analysis

Perform grouping using Pandas:

- Country vs total shipments
- Vendor vs total cost
- Product Group vs total value
- Shipment Mode vs average delay

Task 5: Visualizations (Python)

Create at least 3 charts:

1. Bar chart → Country vs Shipments
2. Pie chart → Shipment Mode distribution
3. Line chart → Delivery trend over time

Part 2: Power BI Dashboard

Dashboard Title:

Supply Chain Performance Dashboard

KPI Cards (Top Section)

Create these KPIs:

- Total Shipments
- Total Freight Cost
- Average Delivery Delay
- Total Line Item Value

Required Visuals

1. Bar Chart

Country vs Total Shipments

Line Chart

Delivery Trend Over Time

Pie Chart

Shipment Mode Distribution

Column Chart

Vendor vs Total Cost



KPI Insight Panel

- Average delay status
- Cost efficiency insight
- Top performing country

Dashboard Rules

- Keep layout clean and simple
- Use consistent colors
- Add proper titles
- Avoid cluttered visuals
- Make dashboard understandable in 5 seconds

Must write:

- 5 insights from Python analysis
- 3 insights from Power BI dashboard
- 2 business recommendations

Important Instructions

1. Your GitHub repository must be public.
2. Your README file must be properly written.
3. Your LinkedIn post must include your GitHub project link and must mention LogicStack.
4. Do not upload incomplete work.
5. Do not copy another intern's README or LinkedIn post.
6. Use your own words in observations.